Parallel LiDARs Meet the Foggy Weather

Yuhang Liu, Yonglin Tian, Boyi Sun, Yiyao Wang, and Fei-Yue Wang

***Abstract*—LiDARs play an important role in autonomous driving. They can provide accurate information about the surroundings, which is essential for perception tasks. Unfortunately, LiDARs' performance will be seriously influenced by adverse weather. Adverse weather is a big challenge to the development of perception systems in autonomous driving. Because of the high cost and time intensity, the lack of LiDAR data under adverse weather is the main problem. Enhanced data is a proper method to solve the data shortage problem. Parallel LiDARs based on parallel theories can provide high-fidelity data through the precise modeling of weather in artificial systems. This paper focuses on Parallel LiDARs in fog. A more precise physical model is used to improve the existing work. The influence of different fog conditions on models is explained with quantitative results in detail. This work is of great help to the targeted improvements of trained models in adverse weather.**

***Index Terms*—Parallel LiDARs, foggy weather, enhanced data.**

I. Introduction

With the rapid development of autonomous driving, LiDAR has gradually become an indispensable sensor in perception systems. It can provide point cloud data with structural information of 3D scenes. With the emergence of deep learning and big datasets of point clouds [1], [2], perception tasks based on point cloud, such as object detection [3], [4], [5] and segmentation [6], [7], have been extensively researched. Although there are a huge amount of data at present, almost all the scenes of collected data are under clear weather. However, data in adverse weather, for example rain, fog, and snow, is more important because adverse weather affects LiDARs' performance seriously. The accuracy of data-driven models trained on current datasets drops significantly in adverse weather [8]. It's necessary to train models with sufficient data in adverse weather but the collection of these data is costly and dangerous in the real world. To solve this problem, Parallel LiDARs based on parallel theory [9] are proposed for generating enhanced data in different weather. Based on the work of [8], [10], we take four types of fog into consideration and provide quantitative results of models influenced by fog. It shows the impact of foggy weather on LiDAR performance clearly and provides help for specific optimization of models.

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The main contribution of this paper is that we use a precise physical model to enhance data with fog in Parallel LiDARs. And the great potential of Parallel LiDARs is discussed.

II. Related Work

A. LiDAR Model

LiDAR model in clear weather has already been established and researched [8], [10]. The overall model is shown in (1). P_R is the power of the received signal, which is calculated by the convolution of the transmitted signal power P_T and impulse response H.

$$P_R(R) = C_A \int_0^{2R/c} P_T(t) H(R - \frac{ct}{2}) dt \tag{1}$$

The transmitted signal is shown in (2). Impulse response H is composed of H_C and H_T in (3). P_0 denotes the peak power of the transmitted signal and τ_H is the half-power pulse width.

$$P_T(t) = \begin{cases} P_0 \sin^2(\frac{\pi}{2\tau_H} t), 0 \le t \le 2\tau_H \\ 0, otherwise \end{cases} \tag{2}$$

$$H(R) = H_C(R) H_T(R) \tag{3}$$

H_C is the part of transmission in impulse response and it can be formulated as (4). $\xi(R)$ denotes the crossover function that is closely related to the coverage of transmitters and receivers. H_T is the part of target reflection in impulse response as shown in (5). R_0 is the distance of target and β_0 is the differential reflectivity of target. $\delta(.)$ means Dirac delta function. The target is often called a hard target in order to distinguish it from a soft target, such as water droplets in the atmosphere.

$$H_C(R) = \frac{\xi(R)}{R^2} \tag{4}$$

$$H_T(R) = H_T^{hard}(R) = \beta_0 \delta(R - R_0) \tag{5}$$

With substituting (2), (3), (4), (5) into (1), LiDAR model can also be formulated in (6).

$$\begin{aligned} P_{R,clear}(R) &= C_A \int_0^{2\xi_H} P_0 \sin^2 \frac{\pi}{2\tau_H} t \frac{1}{R_0^2} \beta_0 \delta(R - \frac{ct}{2} - R_0) dt \\ &= \begin{cases} \frac{C_A P_0 \beta_0}{R_0^2} \sin^2(\frac{\pi(R-R_0)}{c\tau_H}), R_0 \le R \le R_0 + c\tau_H \\ 0, otherwise \end{cases} \end{aligned} \tag{6}$$

B. LiDARs in Fog

The influence of fog on LiDARs has already been researched in [8], [10], [11]. The effect of fog can be regarded as the same as small rain [12]. It can be summarized into two aspects: the attenuation phenomenon of the return signal's power and the emergence of false targets. References [8], [10] has modulated the precise LiDAR model under foggy

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weather and taken both two phenomena into consideration. Reference [8] is a pioneering work that aims at providing enhanced datasets through fog simulation. It provides a new idea for data generation in adverse weather.

Although [8] has proposed a precise LiDAR model in fog, the author has simplified the value of two important parameters, attenuation coefficient, and backscattering coefficient. The Fidelity of enhanced data is affected by this simplification It's necessary to use a more precise model [10] to enhance data in fog. It can provide not only high-fidelity datasets but also quantitative results.

III. Improved LiDAR Model in Fog

Reference [8] is a pioneering work about LiDAR simulation in fog. (4) shows the transmission impulse response in clear weather. It should be formulated as (7) in foggy weather because of the attenuation effect.

$$H_C(R) = \frac{T^2(R))}{R^2}\xi(R) \tag{7}$$

T(R) is the attenuation loss during transmission and is simplified to (8). α is the attenuation coefficient, an important parameter that decides the loss of the return signal's power.

$$T(R) = exp(-\alpha R) \tag{8}$$

False targets are caused by the backscattering of small water droplets in the atmosphere, which are also called soft targets. Impulse response of soft target's reflection can be formulated as (9). β denotes the backscattering coefficient, which is an important parameter in foggy weather. And U means Heaviside function.

$$H_T^{soft}(R) = \beta U(R_0 - R) \tag{9}$$

Impulse response of both hard target and soft target is shown in (10) and the power of the received signal in fog is shown in (11).

$$H(R) = \frac{exp(-2\alpha R\xi(R))}{R^2} \times (\beta U(R_0 - R) + \beta_0 \delta(R - R_0)) \tag{10}$$

$$P_{R,fog}(R) = P_{R,fog}^{hard}(R) + P_{R,fog}^{soft}(R) \tag{11}$$

The received power of a hard target in fog can be calculated by (12). And Simpson's $\frac{1}{3}$ rule is used to get the received power of a soft target.

$$P_{R,fog}^{hard}(R) = exp(-2\alpha R_0)P_{R,clear}(R) \tag{12}$$

With LiDAR model in fog, it's obvious that both the attenuation coefficient and the backscattering coefficient play an important role in modeling. But the value of the attenuation coefficient is determined by approximate estimation and the backscattering coefficient is calculated by visibility simply [8]. It lacks for the support of physical theories. In order to take refinement research about the effect of foggy weather, it's essential to use a precise physical model [10] to calculate these two parameters.

During the transmission of laser beams, scattering at a water droplet in the atmosphere can be regarded as scattering at a liquid sphere. The MIE's theory is suitable for calculating the attenuation efficiency $Q_{EXT}(D)$ and the backscattering efficiency $Q_B(D)$ of each single water droplet. The parameter is closely related to the wavelength of laser beams and the size of water droplets. The most common wavelength of LiDARs used in autonomous driving is 905nm and the corresponding curves of $Q_{EXT}(D)$ and $Q_B(D)$ are shown in Fig. 1.

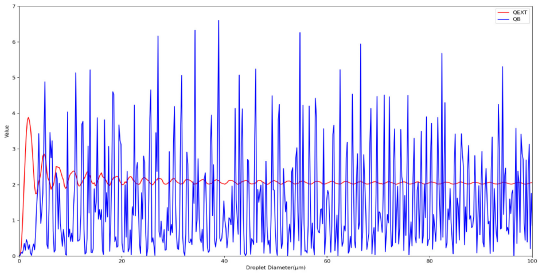


Fig. 1. The curves of $Q_{EXT}(D)$ and $Q_B(D)$

TABLE I
The Parameters for Four Types of Fog

Weather conditions	$\rho(cm^{-3})$	a	γ	$r_C(\mu m)$
Strong advection fog	20	3	1.0	10.00
Moderate advection fog	20	3	1.0	8.00
Strong spray	100	6	1.0	4.00
Moderate spray	100	6	1.0	2.00

TABLE II
Value of the Attenuation Coefficient and the Backscattering Coefficient

Weather conditions	The attenuation coefficient	The backscattering coefficient
Strong advection fog	0.028995	0.020228
Moderate advection fog	0.018727	0.013097
Strong spray	0.017067	0.012338
Moderate spray	0.004486	0.003204

It's obvious that there are numerous water droplets in the atmosphere on the propagation path. A precise model of water droplets' distribution is essential for fog simulation. Gamma distribution is a typical model that is similar to the real scene [10]. We use Gamma distribution to model four types of fog in the atmosphere. The distribution model of water droplets is shown in (13), (14). And the corresponding parameters in different cases of foggy weather are shown in Table I [10].

$$N(D) = \frac{\gamma \rho b^{\frac{a+1}{\gamma}}}{\Gamma(\frac{a+1}{\gamma})}(\frac{D}{2})^a e^{-b(\frac{D}{2})^\gamma} \tag{13}$$

$$b = \frac{a}{\gamma(\frac{D_C}{2})^\gamma} \tag{14}$$

Based on the provided parameters in Table I, the curves of water droplets' distribution in four common fog conditions can be obtained, as shown in Fig. 2.

With this precise model of water droplets' distribution in the atmosphere and the efficiency, the attenuation coefficient and the backscattering coefficient can be formulated as shown in (15), (16).

$$\alpha = \frac{\pi}{8}\int_{D=0}^{\infty} D^2 Q_{EXT}(D)N(D)dD \tag{15}$$

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TABLE III
3D AVERAGE PRECISION (AP) RESULTS ON KITTI DATASET IN DIFFERENT FOG

Weather conditions	PointPillars			SECOND			PointRCNN		
	Car	*Pedestrian*	*Cyclist*	*Car*	*Pedestrian*	*Cyclist*	*Car*	*Pedestrian*	*Cyclist*
Clear weather	77.28	52.29	62.60	78.62	52.98	67.16	78.77	54.40	72.14
Fog simulation in [8]	71.31	36.82	43.28	77.63	45.40	55.64	74.74	48.73	51.90
Strong advection fog	70.78	42.66	40.00	76.69	46.92	47.73	70.33	53.18	51.49
Moderate advection fog	75.59	45.89	52.41	78.11	47.39	58.40	76.30	53.55	61.37
Strong spray	76.34	45.72	54.62	78.34	48.24	58.48	77.78	53.77	67.18
Moderate spray	77.39	52.05	62.30	78.60	51.80	65.63	78.65	55.08	67.02

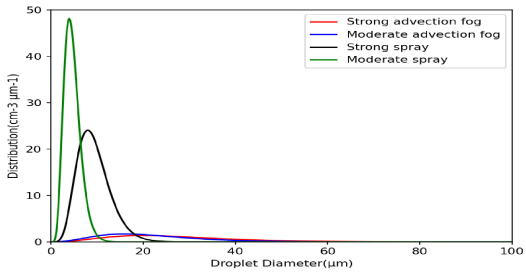


Fig. 2. The distribution of water droplets in the atmosphere

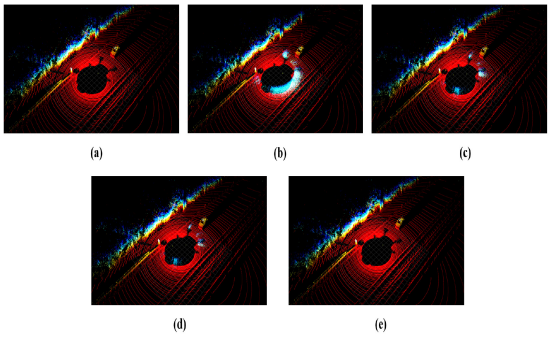


Fig. 3. A scene in clear weather and 4 types of fog.(a) denotes clear weather, (b)-(e)denotes strong/moderate advection fog and spray.

$$\beta = \frac{\pi}{8} \int_{D=0}^{\infty} D^2 Q_B(D) N(D) dD \tag{16}$$

The value of the attenuation coefficient and the backscattering coefficient can be calculated directly using the provided parameters in Table I and (15), (16). In four types of foggy days, the final value of these two parameters is shown in Table II.

With improved LiDAR model in fog, it's obvious that the attenuation coefficient and the backscattering coefficient are closely related to the phenomenons of power attenuation and false targets respectively. The higher the attenuation coefficient, the more severe the power attenuation. And false targets are more likely to appear with a large backscattering coefficient. In clear weather, these two parameters are approximated to 0. However, they can not be simplified to 0 in foggy weather. Table II shows that the influence of strong fog is more serious than moderate fog. And advection fog is a bigger threat to LiDARs compared with spray.

IV. EXPERIMENTS AND DISCUSSIONS

Section III has already explained improved LiDAR model in detail and analyzed the results theoretically. This part focuses on LiDARs' performance influenced by foggy weather with experiments. KITTI dataset [1] is used to generate enhanced data on different foggy weather. Fig. 3 shows a scene of different weather conditions in the KITTI dataset for qualitative analysis.

Fig. 3(b) denotes strong advection fog and it takes the most serious influence on LiDARs compared with other weather conditions. It's more likely to generate false targets around LiDARs. Fig. 3(c), (d) denote moderate advection fog and strong spray respectively. From the figure, we can find that these two types of weather take similar effects to LiDARs. Fig. 3(e) denotes moderate spray and it can be regarded as taking no influence on LiDARs. The qualitative results are consistent with the theoretical analysis in Section III.

3D object detection task is chosen for quantitative analysis of the fog influence. With the help of OpenPCDet [13], 3D object detection models PointPillars [3], SECOND [4], and PointRCNN [5] are used for experiments. The KITTI validation set is processed by improved LiDAR model to generate data in foggy weather. These models are trained based on the original KITTI training set and tested on the enhanced validation set. The results are listed in Table III and also compared with the fog simulation method in [8]. The model performance in four types of fog is consistent with the qualitative analysis. 3D average precision(AP) of models decreases most significantly in strong advection fog. The models in moderate spray can work with only a slight decline in 3D AP. The influence caused by moderate advection fog and strong spray is between the above two fog conditions. In Table III, we also find that the influence on the classes *Pedestrian* and *Cyclist* is more serious than the class *Car*. A proper explanation of this phenomenon is that each pedestrian or cyclist only contains fewer points compared with a car. They are more sensitive to the change in intensity and missing points.

In addition, we choose strong advection fog as an example and divide the influence on point clouds into three categories: intensity change, points missing, and noise points. Each variable is modified during the testing and the results are shown in Table IV. In Table IV, we find that the missing points take the most serious influence on the classes *Car* and *Cyclist*. The effect of intensity change is the most obvious in class *Pedestrian* and noise points can be regarded as making no difference. This is helpful with the specific improvement of models in foggy weather.

Although different fog conditions have already been researched in this work, there are still two problems that need to be solved in the future. The first problem is that

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TABLE IV
COMPARATIVE EXPERIMENTS ABOUT THE INFLUENCE ON POINT CLOUDS CAUSED BY FOG

Variable type	PointPillars			SECOND			PointRCNN		
	Car	*Pedestrian*	*Cyclist*	*Car*	*Pedestrian*	*Cyclist*	*Car*	*Pedestrian*	*Cyclist*
None	77.28	52.29	62.60	78.62	52.98	67.16	78.77	54.40	72.14
Intensity change	76.21	39.59	51.60	78.27	46.05	57.09	78.02	52.35	60.34
Missing points	71.48	50.96	45.59	76.97	50.61	50.20	75.12	54.38	50.98
Noise points	77.13	51.65	61.82	78.57	52.99	66.71	78.72	58.77	71.36

the improved LiDAR model ignores the change in the measurement range. Reference [14] proved that the foggy weather reduces the detection range of LiDARs significantly. However, it's different from the present model, as shown in Fig. 3. It should be optimized in future work. The second problem is that although the appearance of false targets has already been modulated, its principle is different from reality. False targets frequently emerge around LiDARs because the density of laser beams is high in that region [10]. The receivers will get a higher power scattered by water droplets in the atmosphere. A more precise physical model is necessary to solve this problem.

V. CONCLUSION

In this paper, Parallel LiDARs are used to obtain enhanced data on four kinds of fog in cyber space. The qualitative and quantitative results of different fog conditions are analysed. And two problems of the present model are clarified. Parallel LiDARs will help solve more perception tasks in adverse weather because it can help generate a huge amount of enhanced data in artificial systems.

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